COURSE CODE: CSA1412

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO1*: Apply lexical analysis for token specification and recognition using LEX tool (*BL3)

Assessment Tool *Weightage:* 20%

QUESTIONS:

Duration :60 Mins

On class

Total Marks:50

Annexure A

ERROR ANALYSIS TASK

Code *of Conduct*:

I Mathusri *P-(*192311363*)* (Name*/Reg* No) *certify* that this *submission is my original* work *and* that I have *adhered* to the *guidelines specified for* this assessment.

I understand that *any violation* of academic *integrity rules* will *result in disciplinary action*.

Signature of the *student:*

Assignment-2 CSA 1412)

(Compiler design)

0 (a) Error Identification

(Analysis level)

P. Mathusi

192311363

Explanation

invalid identifies variable names can't

variable names can't begin with

a number

miesing semicolon Every

declaration **muut**

Incomed Statement

Error

10

int 2num = 10;

12

intar [5] = {1,2,3,4,$}

16

reuelt

square (); function argument

18

if (*reult* =100)

20

Printf("Revelt $100 \n")

B

total = result +10;

Assignment intend of

comparcion

murring semicolon

undeclared variable

25 Printf ("%d\n", arr[s]] Array

idex out of bonds

28 P=50;

uninitialized pointes

31

int 2=x/y

Division by

чего

35-38 while (i<=5)

infinite loop

end **with**

squared) requires one interger argument

=1

a value

assigns should be cued for

Comparin

printf statement meet

end with..

variable total has not

been declared valid total has not been declared.

pointer poloes not point

to valid memery

y=o; dividing by zewo

Cause

runtime emr

(b) classification of Errors.

Enor

Invalid Indentifier (2num)

missing semicolon after

array

declaration

-Mining function

argument

Compiler Phase

Lexical Error

Syntax Enor

Semantic Error

T

C = =)

Alignment (=) instead of comparison

Logical Error

Missing semicolon after printf

Syntax Emor

undedared variable (total)

Semantic Error

Array index out of bounds
Runtime Emor

unintralized Pointe
Runtime Emor

Divilion by Zeeo
Runtime Error

Infinite loop
Cogical Enor

(0 Compilation Behavior
(Evalution level .)

*Texical Analyses.

-The lexical analyzer defect

s'invalid toben such aus

-invalid identifia 2num

* Syntax Analysis:)

The Parser defects

mixing semicolon after array declaration

-miesing semicolon after printf(

* Sematic analysis.)

The Sematic analyzer defects

Function call with miesing argument

undeclared variable total

* Runtime enors.

There occurs only during execution

-Division by

New

Array idex out of bounds

Dereferencing an unlimited pointer

*Effects

of Earlier Dors

- Lexical Errors stop token generation

12

16

Syntax

exxor prevent pawing

IC

If parsing `fails`, semantic analysis may

not execute

Runtime erwas appear only after successful compilation

(d) Corrected Code

res

```
#include <stdio.h>
print ("%d\n", ;) ;
int
18
square (int n)
i++;
Y
retum n*n;
retumo,
20
y
y
چل
```

25

28

31

35-3

```
int main()
{
int num=co;

}
```

```
int over [s] {1, 2, 3, 4, 5};
int reuelt;
спит) ,
result = square (num)',
if (reult == 100)

printf("Reuelt is 100 ('n  ") ;


int total = reult +10;
```

(b) `printf("%d\n", arr[a]);`

(

```
int value = 50;

*
int + p = &value;
Printf ("%d [n", *P);

int x=20, y = 5;
int2 = x/ys
Print ("%d [n", z);

int i=1;
while (iles)
2
```

→ Division

by

zew: Cause

(e) impact Analys

con

Dereferencing an uninitialized pointer: may

segmentation fault

an uninitialized pointer: may

cause a program crack

a

nentimo

error and terminates the

Program.

-

index out of bounds:

Access es invalid memory

& produces conpredictable reults

Amay

Infinite loop: Program never. Stops & evauto CPU time.

яворя Alignment (=) instead of comparison (==): produces incorrect progrom logic and wrong output.

?

(4) Advanced Compiler Perpective

The Compiler should display clear erros menage with the exact line number and suggestion to help programmen earily identify & fix errors.

2

(a) Errar identification

Line

Error

Reason

int a = 10

missing ;

statement must end with a semicolon.

If Cacb
muring
cloung parentheus is
menin

for Cent i=0;125; `itt`
mixing)
clowing bracket miving

`ahite Caczo`

mining)

cloung pramentheus is

m

Printf ("um = "%d\n", ax+b;

muing)

function call is

incomy

26) <u>Exxar </u>*clarification*

Error

Missing;

I

Mixing **in** if Miring in for

<u>Mining </u>in while

Ming) is printf

10 <u>Syntax </u>Correction.

int a = 10

if (acb) for

```
.(int i=0; i<5; i++)
```

```
while (ac20)
printf ("Sum = "/%d\n",
a+b)
```

(Advanced Thinking:

Grammar Rule:

Category

mising statement teminatx

<u>unmatched </u>parenthesis

incorrect control structure

incorect control structure

Muring delimiter

(d) Compiler Behavour

Analysis

tax errors

The Parser defeds syntax

uring context -Free

Gramman CCFG

- it reports miving

semicolons, braces and parentheus

parser

Error recovery allows the continue finding of Mopping immediatly

more errors instead

if - statement + if

(condition statement.

merriney I Defection.

ти

-The Parser experts a clowing after the condition if it tax error.

mining. `it` reports a *unitar*

to

Two improvement.-

O `Provide` clear errar meuages with linio numbers

C lugest

possible

correction (e.g,

"Dig you

forged "37")

Two improvement.

© Provide clear erros mensages with line numbers C lugject possible correction (e.g, "Dig you forget" 9"7")

(a) identify Emors:

0

Reason

is not *allowed* variable

name

statement must end with

a semicolon

square() requires *one*

Line

Errons

int #count=5;

invalid identifies

Mening;

float rate=25

nevalt squares) ;

mining argument

if (revelt >*20*)

mening 3 before che

syntax

int z = x/y;

=

Diverion by Array out

of bounds

zero

y=o cause nentime

marks [4]

incorrect if-else

error

G

valid index is 0-3

*ptr =25;

Meninitialized

pointer

pointer is not initialized

total = x +0+y;

while(i<=5)

Redundant operation

infinite loop

to is unecessary

i is never incremented

6 classify by

compila phone:

*

* lexical Excx: invalid

identifier (#courd)

Syntax Error:
mining
semicolon, mining
brace * Semantic
Error: Mining
function argument.
*Intermediate
cate eine invalid

functions call.

* Optimization imme: total =

x+0+y * Runtime Errar:

Dévérios by pointer

zero, a

• Compilation vs

Execution Analyur

*

Lexical: invalid identifier

array

out

of bounds, curintialirad

Syntax : Mening

semicolon & mixing brace.

* Semantic : Mering

frenetion argument.

* Runtime: Diverion by

* Texical and Syntax

d) Program Correction

кат

*

2010, invaliol

errars

Replace #count with count

* Add mining semicolon

call square (count).

*

+ correct the if-else

braces

* use y=5 before divirion *

Access markas [3]

invaliol pointe▪ array

out of bounch

may stop later complice

phones.

* Initilize pointer before dereferoning

* Remove `+0`.

*Add i++ inside the loop.

e) Deep Analyus

* unintialized <u>pointer </u>:
may crack the program *
Am
Array out of
bounds: Acesses
invalid memary
* In
Infinite Cap: Program
+ Devein by?
+
never terminate

Telo:– causes runtime euor.

Mining function argument: aures compilation facture.

ident

RUBRICS FOR CALCULATION:

Level 4
Criteria
Weightage
Level 3 (Proficient)

Error
(Exemplary)

Accurately identifies

all errors **with** clear
20**%**
Identification
understanding
of
Identifies most **errors**

with minor omissions
Level 2

(Developing)

Identifies limited
Level **1**

(Beginning)

Fails to identify
errors； some are

missed
key errors

issues

Clearly explains
Unable
to
Root Cause
30**%**
Analysis
underlying causes

with

strong
Explains causes with

moderate clarity
Partial explanation;

weak reasoning
identify causes

of errors
technical reasoning

Correction &
25%
Justification
Provides correct

solutions
with

strong justification
Provides

corrections with

some justification
correct
Corrections are
Incorrect or **no**

partially correct or corrections

weakly justified
provided
and logic

Effectively applies

Application
relevant concepts to
Applies
concepts
Limited application;
Unable
to
15%
of Concepts
analyze and resolve
with minor errors
requires guidance
apply concepts

errors

Poor

Clarity &

Presentation
10%
Presents analysis

clearly,
logically, and systematically
Generally clear with Lacks clarity or

minor issues
presentation

proper structure
and difficult to

understand